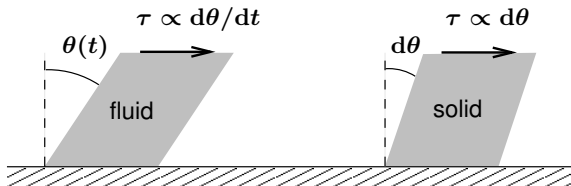


Introductory remarks and starting assumptions

Solids, liquids and gases

- Solids:**
- under the action of a small external force, solids respond with a small deformation
 - internal force counteracting the internal force is proportional to the deformation
 - $\rightsquigarrow$ fixed shape

- Fluids:**
- easily deformed under the action of external forces
 - internal resistive force is proportional to the rate of deformation
 - $\rightsquigarrow$ adapt their shape to that of the container



- Liquids:**
- high density
 - very low compressibility
 - $\rightsquigarrow$ fixed volume, free surfaces

Density: $\rho_{\text{liquid}} \gg \rho_{\text{gas}}$

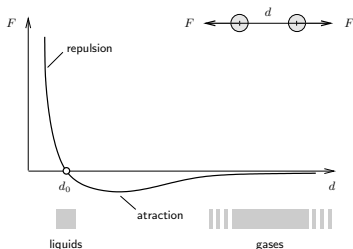
- Gases:**
- low density
 - high compressibility
 - $\rightsquigarrow$ use all the volume available

Compressibility:

$$\left(\frac{\partial \rho}{\partial p} \right)_{T, \text{liquid}} \ll \left(\frac{\partial \rho}{\partial p} \right)_{T, \text{gas}}$$

Solids, liquids and gases

The differences in macroscopic properties are the result of different microscopic structure.



F intermolecular force

d intermolecular distance

d_0 characteristic molecular size $\sim 3 \cdot 10^{-10} \text{ m}$

ρ density

N_A Avogadro's number, $6.022 \cdot 10^{23}$ molecules/mol

W molar mass

The intermolecular distance d can be estimated by

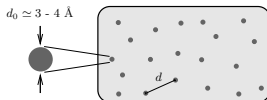
$$\rho = \frac{W/N_A}{d^3} \equiv \frac{\text{mass of 1 molecule}}{\text{volume corresponding to 1 molecule}} \rightarrow d = \left(\frac{W}{\rho N_A} \right)^{1/3}$$

Gases ($d \simeq 10 d_0$)

Low density

High compressibility

Use all the volume available



Air:

$$\rho = 1.2 \text{ kg/m}^3 \quad W = 28.96 \text{ g/mol}$$

$$\leadsto d \simeq 3.4 \cdot 10^{-9} \text{ m}$$

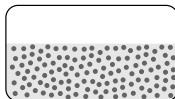
$$\leadsto 10^{16} \text{ molecules in } 1 \text{ mm}^3$$

Liquids ($d \simeq d_0$)

High density

Low compressibility

Form free surfaces



Water:

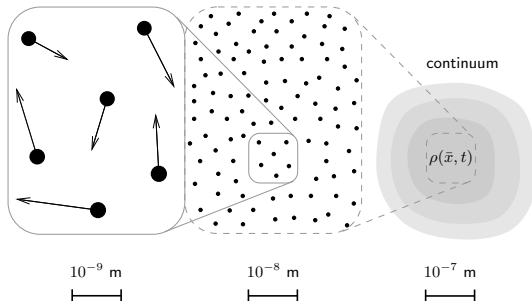
$$\rho = 10^3 \text{ kg/m}^3 \quad W = 18.02 \text{ g/mol}$$

$$\leadsto d \simeq 3.1 \cdot 10^{-10} \text{ m}$$

$$\leadsto 10^{19} \text{ molecules en } 1 \text{ mm}^3$$

The fluid as a continuum: fluid particle

The **continuum hypothesis** allows the definition of a range of scales where the fluid properties can be described as continuous functions of position and time.



Fluid particle: small volume of fluid δV with respect to which we shall define the concepts of density, velocity, etc. Each particle is centered at a location $\vec{x}$ at a given time t and its size must be:

- sufficiently large to allow averaging the fluid properties in it and
- sufficiently small so it can be considered a material point at $\vec{x}$.

The knowledge of the properties of each fluid particle at a given instant of time will provide a precise description of the fluid field (density, velocity, etc.) as a function of position $\vec{x}$ and time t .

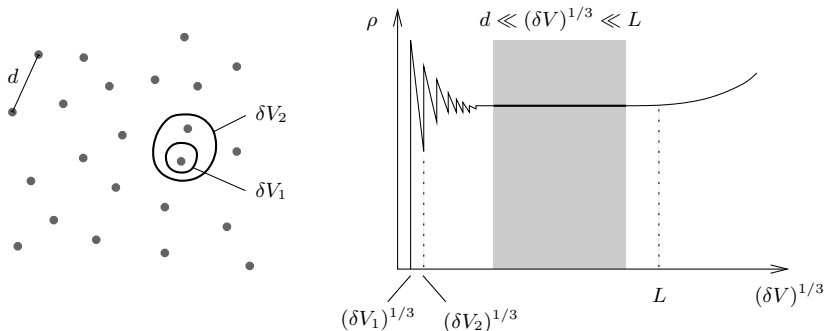
Fluid particle

- d intermolecular distance, $d \simeq 3.4 \cdot 10^{-9}$ m for air at nc and $d \simeq 3.1 \cdot 10^{-10}$ m for water
- L macroscopic characteristic distance over which significant variations of flow properties are observed

Lets compute the density of a fluid particle of volume δV according to

$$\rho = \frac{\Sigma m_i}{\delta V},$$

where Σm_i is the mass of all molecules found inside the volume considered:



Fluid particle: fluid volume δV centered at $\bar{x}$ at the instant t , $d \ll (\delta V)^{1/3} \ll L$

Density, velocity and internal energy

d intermolecular distance, $d \simeq 3.4 \cdot 10^{-9}$ m for air and $d \simeq 3.1 \cdot 10^{-10}$ m for water

L macroscopic characteristic length

m_i mass of the molecule i

$\bar{v}_i$ velocity of the molecule i

E_i total energy of the molecule i

Fluid particle: fluid volume δV centered at $\bar{x}$ at the instant t , $d \ll (\delta V)^{1/3} \ll L$

Density:
$$\rho(\bar{x}, t) = \lim_{\delta V \rightarrow 0} \frac{\sum m_i}{\delta V}$$

Velocity:
$$\bar{v}(\bar{x}, t) = \lim_{\delta V \rightarrow 0} \frac{\sum m_i \bar{v}_i}{\sum m_i}$$

Internal energy p.u. mass:
$$e(\bar{x}, t) = \lim_{\delta V \rightarrow 0} \frac{\sum E_i}{\sum m_i} - \frac{|\bar{v}|^2}{2}$$

Local thermodynamic equilibrium

Classical thermodynamics

Homogeneous system in mechanical and thermal equilibrium, with properties uniform in $\vec{x}$ and constant in t : the thermodynamic state is completely determined by the values of any two independent state variables, any other variable can be computed through the equations of state.

Fluid mechanics deals with systems that are not in thermodynamic equilibrium, with fluid variables changing both spatially and temporally. However, the results of equilibrium thermodynamics are approximately valid for most of the non-equilibrium states found in fluid mechanics.

By the [local thermodynamic equilibrium hypothesis](#), an observer moving with the local fluid velocity can describe the fluid state in terms of thermodynamics variables that are inter-related by the same equations of state that apply for systems in strict equilibrium. It is as if, at every instant of time, each fluid particle is at the equilibrium state corresponding to the local density and internal energy values.

- Rigorously justified through the kinetic theory for gases
- Substantiated by ample experimental evidence for liquids

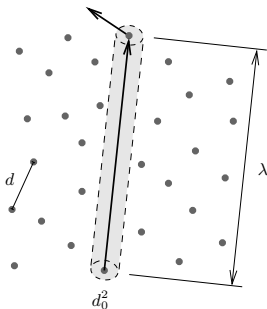
Local thermodynamic equilibrium

Molecular collisions, whereby molecules exchange momentum and energy and thus adjust themselves to the local state of molecular motion, constitute the mechanism that enables thermodynamic equilibrium to be attained in a gas. For gas systems in which the distance travelled by a molecule between two subsequent collisions, called *mean free path* λ , is much smaller than the macroscopic length L , each molecule will undergo a large number of collisions before reaching regions with different macroscopic properties.

The mean free path λ can be estimated by equating the volume swept by the molecule in its displacement between collisions, $d_0^2 \lambda$, to the volume per molecule that exists in the gas, d^3 ,

$$\lambda/d \simeq (d/d_0)^2.$$

For air at normal conditions, $\lambda \sim 4 \cdot 10^{-7} \text{ m} \sim 128d$.



Local thermodynamic equilibrium

$$\frac{\lambda}{L} \ll 1$$

Continuum

$$\frac{d}{L} \ll 1$$

The criterion for local thermodynamic equilibrium hypothesis is more restrictive than that for the continuum hypothesis, thus we define a **fluid particle** with the condition

$$\lambda \ll (\delta V)^{1/3} \ll L$$

The local thermodynamic equilibrium allows to apply locally the thermodynamics laws and relations.

Thermodynamic variables and useful relationships

With $\bar{v}(\bar{x}, t)$ and two state variables, e.g. $\rho(\bar{x}, t)$ and $e(\bar{x}, t)$ (already defined), the local state of the fluid is fully determined. The remaining thermodynamic variables are determined through the equations of state.

Entropy

$$s = s(e, \rho) \quad \text{or} \quad e = e(s, \rho)$$

$$de = T ds - p d\left(\frac{1}{\rho}\right)$$

Enthalpy

$$h = e + \frac{p}{\rho}$$

Temperature

$$T = \left(\frac{\partial e}{\partial s}\right)_{\rho}$$

Pressure

$$p = -\left(\frac{\partial e}{\partial \rho^{-1}}\right)_s$$

c specific heat for a liquid

$R_g = R^{\circ}/W$ specific gas constant

$R^{\circ} = 8.314 \text{ J/(mol K)}$ universal gas constant

W molar mass of the gas

c_v specific heat at constant volume

c_p specific heat at constant pressure

$\gamma = c_p/c_v$ ratio of specific heats for a gas

Perfect liquid (constant ρ, c)

$$\rho = \rho_0$$

$$e = cT + e_0$$

$$h = cT + e_0 + \frac{p}{\rho_0}$$

$$s = c \ln(T) + s_0$$

Perfect gas (constant c_v, c_p)

$$p = \rho R_g T$$

$$c_p = c_v + R_g$$

$$e = c_v T + e_0 = \frac{1}{\gamma-1} \frac{p}{\rho} + e_0$$

$$h = c_p T + e_0 = \frac{\gamma}{\gamma-1} \frac{p}{\rho} + e_0$$

$$s = c_v \ln\left(\frac{p}{\rho^{\gamma}}\right) + s_0$$

Water

$$\rho_0 = 10^3 \text{ kg/m}^3$$

$$c = 4180 \text{ J/(kg K)}$$

Air at normal conditions

$$R_g = 287 \text{ J/(kg K)}$$

$$c_v = 717 \text{ J/(kg K)}, c_p = 1004 \text{ J/(kg K)}, \gamma = 1.4$$

Kinematics

Coordinate systems

Orthogonal curvilinear coordinates with local unit vectors ($\bar{e}_1, \bar{e}_2, \bar{e}_3$)

Cartesian

$$x_1 = x, x_2 = y, x_3 = z \\ d\bar{x} = (dx, dy, dz)$$

Cylindrical

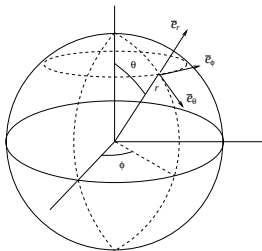
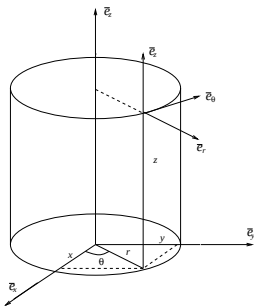
$$x_1 = r, x_2 = \theta, x_3 = z \\ d\bar{x} = (dr, r d\theta, dz)$$

Spherical

$$x_1 = r, x_2 = \theta, x_3 = \phi \\ d\bar{x} = (dr, r d\theta, r \sin\theta d\phi)$$

scale factors

$$d\bar{x} = (h_1 dx_1, h_2 dx_2, h_3 dx_3)$$



Cartesian:

$$h_1 = h_2 = h_3 = 1$$

Cylindrical:

$$h_1 = 1, h_2 = r, h_3 = 1$$

Spherical:

$$h_1 = 1, h_2 = r, h_3 = r \sin\theta$$

Vector and tensor formulae

gradient of a scalar function

$$\nabla \Phi = \left(\frac{1}{h_1} \frac{\partial \Phi}{\partial x_1}, \frac{1}{h_2} \frac{\partial \Phi}{\partial x_2}, \frac{1}{h_3} \frac{\partial \Phi}{\partial x_3} \right)$$

laplacian of a scalar function

$$\nabla^2 \Phi = \frac{1}{h_1 h_2 h_3} \left[\frac{\partial}{\partial x_1} \left(\frac{h_2 h_3}{h_1} \frac{\partial \Phi}{\partial x_1} \right) + \frac{\partial}{\partial x_2} \left(\frac{h_1 h_3}{h_2} \frac{\partial \Phi}{\partial x_2} \right) + \frac{\partial}{\partial x_3} \left(\frac{h_1 h_2}{h_3} \frac{\partial \Phi}{\partial x_3} \right) \right]$$

divergence of a vector function

$$\nabla \cdot \bar{a} = \frac{1}{h_1 h_2 h_3} \left[\frac{\partial}{\partial x_1} (h_2 h_3 a_1) + \frac{\partial}{\partial x_2} (h_1 h_3 a_2) + \frac{\partial}{\partial x_3} (h_1 h_2 a_3) \right]$$

curl of a vector function

$$\begin{aligned} \nabla \wedge \bar{a} &= \frac{1}{h_1 h_2 h_3} \begin{vmatrix} h_1 \bar{e}_1 & h_2 \bar{e}_2 & h_3 \bar{e}_3 \\ \frac{\partial}{\partial x_1} & \frac{\partial}{\partial x_2} & \frac{\partial}{\partial x_3} \\ h_1 a_1 & h_2 a_2 & h_3 a_3 \end{vmatrix} = \frac{1}{h_2 h_3} \left[\frac{\partial}{\partial x_2} (h_3 a_3) - \frac{\partial}{\partial x_3} (h_2 a_2) \right] \bar{e}_1 + \\ &\quad \frac{1}{h_1 h_3} \left[\frac{\partial}{\partial x_3} (h_1 a_1) - \frac{\partial}{\partial x_1} (h_3 a_3) \right] \bar{e}_2 + \frac{1}{h_1 h_2} \left[\frac{\partial}{\partial x_1} (h_2 a_2) - \frac{\partial}{\partial x_2} (h_1 a_1) \right] \bar{e}_3 \end{aligned}$$

Vector and tensor formulae

gradient of a scalar function, in cartesian coordinates

$$\nabla \Phi = \frac{\partial \Phi}{\partial x} \bar{e}_x + \frac{\partial \Phi}{\partial y} \bar{e}_y + \frac{\partial \Phi}{\partial z} \bar{e}_z$$

laplacian of a scalar function, in cartesian coordinates

$$\nabla^2 \Phi = \frac{\partial^2 \Phi}{\partial x^2} + \frac{\partial^2 \Phi}{\partial y^2} + \frac{\partial^2 \Phi}{\partial z^2}$$

divergence of a vector function, in cartesian coordinates

$$\nabla \cdot \bar{a} = \frac{\partial a_x}{\partial x} + \frac{\partial a_y}{\partial y} + \frac{\partial a_z}{\partial z}$$

curl of a vector function, in cartesian coordinates

$$\nabla \wedge \bar{a} = \begin{vmatrix} \bar{e}_x & \bar{e}_y & \bar{e}_z \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ a_x & a_y & a_z \end{vmatrix} = \left(\frac{\partial a_z}{\partial y} - \frac{\partial a_y}{\partial z} \right) \bar{e}_x + \left(\frac{\partial a_x}{\partial z} - \frac{\partial a_z}{\partial x} \right) \bar{e}_y + \left(\frac{\partial a_y}{\partial x} - \frac{\partial a_x}{\partial y} \right) \bar{e}_z$$

Vector and tensor formulae

gradient of a vector function

$$(\nabla \bar{a})_{ii} = \frac{1}{h_i} \frac{\partial a_i}{\partial x_i} + \sum_{k \neq i} \frac{a_k}{h_i h_k} \frac{\partial h_i}{\partial x_k}$$

$$(\nabla \bar{a})_{ij} = \frac{1}{h_i} \frac{\partial a_j}{\partial x_i} - \frac{a_i}{h_i h_j} \frac{\partial h_i}{\partial x_j} \quad \text{if } i \neq j$$

divergence of a tensor

$$(\nabla \cdot \bar{\bar{A}})_i = \frac{h_i}{h} \sum_j \frac{\partial}{\partial x_j} \left(\frac{h A_{ij}}{h_i h_j} \right) + \sum_j \frac{A_{ij} + A_{ji}}{h_i h_j} \frac{\partial h_i}{\partial x_j} - \sum_j \frac{A_{jj}}{h_i h_j} \frac{\partial h_j}{\partial x_i}$$

where $h = h_1 h_2 h_3$.

Vector and tensor formulae

gradient of a vector function, in cartesian coordinates

$$\nabla \bar{a} = \begin{bmatrix} \frac{\partial a_x}{\partial x} & \frac{\partial a_y}{\partial x} & \frac{\partial a_z}{\partial x} \\ \frac{\partial a_x}{\partial y} & \frac{\partial a_y}{\partial y} & \frac{\partial a_z}{\partial y} \\ \frac{\partial a_x}{\partial z} & \frac{\partial a_y}{\partial z} & \frac{\partial a_z}{\partial z} \end{bmatrix}$$

divergence of a tensor, in cartesian coordinates

$$\begin{aligned} \nabla \cdot \bar{\bar{A}} = & \left(\frac{\partial A_{xx}}{\partial x} + \frac{\partial A_{xy}}{\partial y} + \frac{\partial A_{xz}}{\partial z} \right) \bar{e}_x + \\ & \left(\frac{\partial A_{yx}}{\partial x} + \frac{\partial A_{yy}}{\partial y} + \frac{\partial A_{yz}}{\partial z} \right) \bar{e}_y + \\ & \left(\frac{\partial A_{zx}}{\partial x} + \frac{\partial A_{zy}}{\partial y} + \frac{\partial A_{zz}}{\partial z} \right) \bar{e}_z \end{aligned}$$

Eulerian and Lagrangian descriptions

Two different approaches to describe the velocity field:

Lagrangian Takes a discrete point of view by treating the fluid as an ensemble of fluid particles. The flow field is described in terms of the fluid-particle trajectories

$$\bar{x} = \bar{x}_T(\bar{x}_o, t)$$

which are a function of their initial position $\bar{x}_o$ and time t . The velocity and acceleration are computed by straightforward derivation

$$\bar{v} = \frac{\partial \bar{x}_T}{\partial t} \qquad \bar{a} = \frac{\partial^2 \bar{x}_T}{\partial t^2}$$

Eulerian Assumes the velocity to be a continuous vector field $\bar{v}(\bar{x}, t)$ as a function of the position $\bar{x}$ and time t .

We shall adopt the Eulerian description

Uniform flow; steady flow; stagnation points

Uniform flow when the spatial differences of the different flow properties are strictly zero.

A uniform velocity field satisfies $\bar{v} = \bar{v}(t)$.

Steady flow when it does not change in time.

A steady velocity field is such that $\bar{v} = \bar{v}(\bar{x})$.

The steadiness of a given flow depends on the observer, for instance, when considering the flow around a moving object.

Stagnation point A point where the velocity is zero (all three components), $\bar{v}(\bar{x}, t) = 0$.

The location of the stagnation points depends on the reference frame selected.

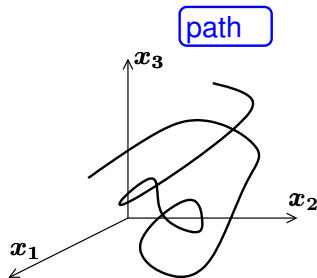
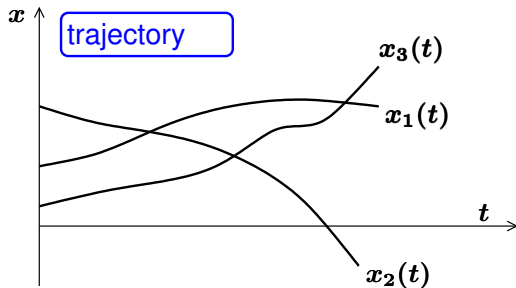
Trajectories and path lines

In an eulerian description of the flow, given a velocity field $\bar{v}(\bar{x}, t)$, the location of the fluid particle at each instant of time is determined by integration of:

$$\left. \begin{aligned} \frac{d\bar{x}}{dt} &= \bar{v}(\bar{x}, t) \\ \bar{x}(t = t_o) &= \bar{x}_o \end{aligned} \right\} \rightarrow \bar{x} = \bar{x}_T(\bar{x}_o; t)$$

The **trajectory** contains information about the path followed by the particle and also about the rate with which it travels along it.

The equations that describe the **path lines** can be obtained by eliminating the time t in the expression for the trajectory.



Fluid lines, fluid surfaces and fluid volumes

fluid line

If $\bar{v}(\bar{x}, t)$ is continuous, fluid particles initially located along the curve $\bar{x} = \bar{x}_l(\lambda)$ (λ denotes a parameter describing the curve) will continue to form a curve at any subsequent time:

$$\bar{x} = \bar{x}_T(\bar{x}_l(\lambda), t)$$

fluid surface

Similarly, fluid particles forming initially a surface $\bar{x} = \bar{x}_s(\alpha, \beta)$ (α and β are parameters) will continue to form a surface:

$$\bar{x} = \bar{x}_T(\bar{x}_s(\alpha, \beta), t)$$

fluid volume

If the surface is initially closed, it can be expected to remain closed in its evolution. The finite volume of fluid bounded by a closed fluid surface is called a fluid volume.

It is clear that mass cannot cross a fluid surface, which implies that the mass of a fluid volume remains constant.

Stream lines, stream surfaces and stream tubes

stream lines

lines that are tangent to the local velocity vector at each point:

$$\frac{h_1 dx_1}{v_1(x_1, x_2, x_3, t)} = \frac{h_2 dx_2}{v_2(x_1, x_2, x_3, t)} = \frac{h_3 dx_3}{v_3(x_1, x_2, x_3, t)}$$

The two integration constants for the above pair of differential equations serve to identify a specific stream line.

Streamlines can only intersect at stagnation points.

For a steady velocity field, path lines and stream lines coincide.

cartesian

$$\frac{dx}{v_x} = \frac{dy}{v_y} = \frac{dz}{v_z}$$

cylindrical

$$\frac{dr}{v_r} = \frac{r d\theta}{v_\theta} = \frac{dz}{v_z}$$

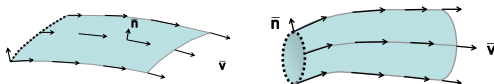
spherical

$$\frac{dr}{v_r} = \frac{r d\theta}{v_\theta} = \frac{r \sin \theta d\phi}{v_\phi}$$

stream surface

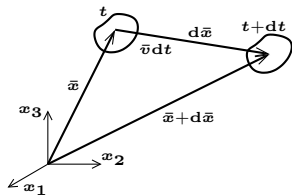
Surface formed by all streamlines intersecting a given arbitrary line.

If the line selected is closed, the resulting stream surface would form a **stream tube**.



Material derivative

To determine the variation of any intensive scalar property ϕ (such as density, pressure or temperature) following the fluid flow (that is, following a fluid particle in its motion), one must account for the fluid motion by considering the infinitesimal displacement of a fluid particle:



At time t the fluid particle is located at $\bar{x}$, so that the initial value of the intensive property is given by $\phi(\bar{x}, t)$. At time $t + dt$, the particle has moved to occupy a new position given by $\bar{x} + d\bar{x}$ (with $d\bar{x} = \bar{v} dt$), and the new value of ϕ would be given by $\phi(\bar{x} + d\bar{x}, t + dt)$.

The variation of ϕ for the fluid particle can be written in the approximate form

$$d\phi = \phi(\bar{x} + d\bar{x}, t + dt) - \phi(\bar{x}, t) = dt \frac{\partial \phi}{\partial t} + d\bar{x} \cdot \nabla \phi$$

Dividing the above expression by dt and identifying $\bar{v} = \frac{d\bar{x}}{dt}$ leads

$$\frac{D\phi}{Dt} = \frac{\partial \phi}{\partial t} + \bar{v} \cdot \nabla \phi,$$

material derivative

$$\frac{D(\quad)}{Dt} = \underbrace{\frac{\partial(\quad)}{\partial t}}_{\text{local temporal variation}} + \underbrace{\bar{v} \cdot \nabla(\quad)}_{\text{convective derivative}}$$

Acceleration of a fluid particle

The concept of material derivative can be extended to determine the time evolution of a vector property.

acceleration vector

$$\bar{a} = \frac{D\bar{v}}{Dt} = \frac{\partial \bar{v}}{\partial t} + \bar{v} \cdot (\nabla \bar{v}),$$

where the gradient of velocity $\nabla \bar{v}$ is a tensor.

cartesian coordinates

$$(\nabla \bar{v})_{ij} = \partial v_j / \partial x_i$$

each component of the acceleration vector a_i reduces to the material derivative of the corresponding velocity component according to

$$a_i = \frac{Dv_i}{Dt} = \frac{\partial v_i}{\partial t} + \sum_j v_j \frac{\partial v_i}{\partial x_j}$$

Note that this last result does not hold when cylindrical or spherical coordinates are used.

A convenient way to express the acceleration, valid regardless of the coordinate system selected,

$$\bar{a} = \frac{\partial \bar{v}}{\partial t} + \nabla \left(\frac{|\bar{v}|^2}{2} \right) - \bar{v} \wedge (\nabla \wedge \bar{v})$$

Circulation and vorticity

circulation along a curve

$$\Gamma = \int_L \bar{\mathbf{v}} \cdot d\bar{\mathbf{l}}$$

where the vector $d\bar{\mathbf{l}}$ represents the differential line element along the curve L

- Gives a measure of the fluid motion along the direction of the curve
- If $\bar{\mathbf{x}} = \bar{\mathbf{x}}_l(\lambda)$ ($\lambda_1 < \lambda < \lambda_2$) is the parametric representation of the curve, then

$$d\bar{\mathbf{l}} = \frac{d\bar{\mathbf{x}}_l}{d\lambda} d\lambda \quad \text{and} \quad \Gamma = \int_L \bar{\mathbf{v}}(\bar{\mathbf{x}}, t) \cdot d\bar{\mathbf{l}} = \int_{\lambda_1}^{\lambda_2} \bar{\mathbf{v}}(\bar{\mathbf{x}}_l(\lambda), t) \cdot \frac{d\bar{\mathbf{x}}_l}{d\lambda} d\lambda$$

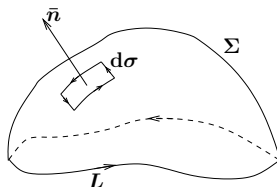
- If one considers a closed curve and $\bar{\mathbf{v}}(\bar{\mathbf{x}}, t)$ is a continuous function, then, according to Stokes theorem,

$$\Gamma = \oint_L \bar{\mathbf{v}} \cdot d\bar{\mathbf{l}} = \int_{\Sigma} (\nabla \wedge \bar{\mathbf{v}}) \cdot \bar{\mathbf{n}} d\sigma$$

Σ represents any surface bounded by L completely contained within the fluid domain

$d\sigma$ is a surface differential element

$\bar{\mathbf{n}}$ unit vector normal to $d\sigma$, defined for the surface to have positive orientation according to the so-called *right-hand rule*



vorticity

$$\bar{\omega} = \nabla \wedge \bar{\mathbf{v}}$$

Irrotational flow and velocity potential

irrotational flow

$\bar{\omega} = \nabla \wedge \bar{v} = 0$, vorticity is identically zero everywhere

by Stokes' theorem, $\Gamma = \oint_L \bar{v} \cdot d\vec{l} = \int_{\Sigma} \bar{\omega} \cdot \bar{n} d\sigma$

- If the circulation along any closed curve is zero, the flow is necessarily irrotational. To demonstrate this consider an infinitesimally small closed curve around a given point

$$\Gamma = \bar{\omega} \cdot \bar{n} d\sigma = 0 \quad \Rightarrow \quad \bar{\omega} = 0$$

- The opposite is not necessarily true, that is, even though $\nabla \wedge \bar{v} = 0$ everywhere in the flow field, the circulation around a closed curve might not be zero. This is so when we cannot define a surface bounded by the curve that is completely contained in the fluid domain. For example, in 2D flows around an object.
- An irrotational flow can be derived from a scalar potential function

velocity potential function

$$\bar{v} = \nabla \varphi$$

- Constant of integration is arbitrary
- Important simplification in the description of irrotational flows: the computation of a three-component vector field (the velocity $\bar{v}$) can be replaced by that of a scalar field (the potential φ).

Convective flux

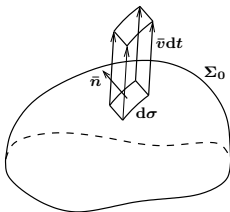
Mass, momentum and energy are transported by the moving fluid. Considering an imaginary fixed surface, the fluid will cross the surface in its motion, transporting across mass, momentum and energy. To quantify in general this transport rate,

$\phi(\vec{x}, t)$: represents a given fluid magnitude expressed per unit volume, with particular cases of interest being the mass, momentum and energy

$\int_V \phi \, dV$: total amount of that given quantity existing within a given volume V

Convective flux of an extensive fluid magnitude

Amount of that fluid magnitude that crosses a surface per unit time



Σ_0 : fixed arbitrary surface

$d\sigma$: surface element oriented perpendicular to $\vec{n}$

$(\vec{v} \cdot \vec{n}) \, d\sigma \, dt$: volume of fluid crossing in a time dt through the differential surface element $d\sigma$ (volume of the parallelepiped shown in the figure)

$\phi (\vec{v} \cdot \vec{n}) \, d\sigma \, dt$: amount of the fluid magnitude ϕ that crosses the surface element $d\sigma$ with the fluid in a time dt

$\phi (\vec{v} \cdot \vec{n}) \, d\sigma$: amount of the fluid magnitude ϕ that crosses the surface element $d\sigma$ with the fluid per unit time

$\int_{\Sigma_0} \phi (\vec{v} \cdot \vec{n}) \, d\sigma$: total amount of the fluid magnitude ϕ that crosses the surface Σ_0 with the fluid per unit time

Convective flux

Amount of the fluid magnitude ϕ that crosses the surface Σ_0 with the fluid per unit time : $\int_{\Sigma_0} \phi \bar{\mathbf{v}} \cdot \bar{\mathbf{n}} \, d\sigma$

fluid magnitude	fluid magnitude p.u.volume, ϕ	flux p.u.surface, $\phi \bar{\mathbf{v}}$
volume	$\phi = 1$	$\phi \bar{\mathbf{v}} = \bar{\mathbf{v}}$: volume flux vector
mass	$\phi = \rho$	$\phi \bar{\mathbf{v}} = \rho \bar{\mathbf{v}}$: mass flux vector
momentum	$\phi \rightsquigarrow \rho \bar{\mathbf{v}}$	$\phi \bar{\mathbf{v}} \rightsquigarrow \rho \bar{\mathbf{v}} \bar{\mathbf{v}}$: momentum flux tensor
energy	$\phi = \rho (e + v^2/2)$	$\phi \bar{\mathbf{v}} = \rho (e + v^2/2) \bar{\mathbf{v}}$: energy flux vector

Convective flux through a moving surface

The concept of convective flux can be extended to a moving surface $\Sigma_c(t)$ whose points are moving with velocity $\bar{\mathbf{v}}_c(\bar{\mathbf{x}}, t)$ by replacing the fluid velocity $\bar{\mathbf{v}}$ with the relative velocity $\bar{\mathbf{v}} - \bar{\mathbf{v}}_c$ when computing the flux to give

$$\int_{\Sigma_c(t)} \phi [(\bar{\mathbf{v}} - \bar{\mathbf{v}}_c) \cdot \bar{\mathbf{n}}] \, d\sigma$$

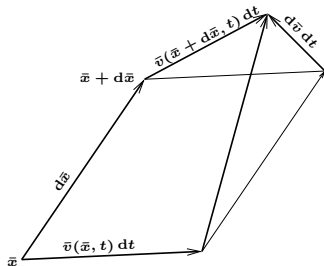
Relative motion near a point

Remember: the force exerted by a portion of fluid on an adjacent portion of fluid is proportional to the rate at which the fluid is being deformed

objective

Determine the local deformation rate in a fluid velocity field $\bar{v}(\bar{x}, t)$

consider the motion of a differential element of fluid line $d\bar{x}$



new length of the line element at $t + dt$:

$$\begin{aligned} & \bar{x} + d\bar{x} + \bar{v}(\bar{x} + d\bar{x}, t)dt - [\bar{x} + \bar{v}(\bar{x}, t)dt] = \\ & d\bar{x} + \underbrace{\bar{v}(\bar{x} + d\bar{x}, t)dt - \bar{v}(\bar{x}, t)dt}_{\substack{d\bar{v}dt \\ \text{(distorsion)}}} \end{aligned}$$

in a first approximation, $d\bar{v} = d\bar{x} \cdot \nabla \bar{v}$

Relative motion near a point

The geometrical character of $d\bar{v} \, dt = d\bar{x} \cdot \nabla \bar{v} \, dt$, is best investigated by decomposing the velocity gradient tensor $\nabla \bar{v}$ into parts that are symmetrical, $\bar{\bar{T}}_d$, and anti-symmetrical, $\bar{\bar{T}}_r$

$$\nabla \bar{v} = \underbrace{\frac{1}{2}(\nabla \bar{v} + \nabla \bar{v}^T)}_{\substack{\bar{\bar{T}}_d \\ \text{(symmetric)} \\ \text{(rate-of-strain tensor)}}} + \underbrace{\frac{1}{2}(\nabla \bar{v} - \nabla \bar{v}^T)}_{\substack{\bar{\bar{T}}_r \\ \text{(anti-symmetric)} \\ \text{(rate-of-rotation tensor)}}$$

cartesian coordinates

$$\bar{\bar{T}}_d = \frac{1}{2} \begin{bmatrix} 2 \frac{\partial v_1}{\partial x_1} & \left(\frac{\partial v_2}{\partial x_1} + \frac{\partial v_1}{\partial x_2} \right) & \left(\frac{\partial v_3}{\partial x_1} + \frac{\partial v_1}{\partial x_3} \right) \\ \left(\frac{\partial v_2}{\partial x_1} + \frac{\partial v_1}{\partial x_2} \right) & 2 \frac{\partial v_2}{\partial x_2} & \left(\frac{\partial v_3}{\partial x_2} + \frac{\partial v_2}{\partial x_3} \right) \\ \left(\frac{\partial v_3}{\partial x_1} + \frac{\partial v_1}{\partial x_3} \right) & \left(\frac{\partial v_3}{\partial x_2} + \frac{\partial v_2}{\partial x_3} \right) & 2 \frac{\partial v_3}{\partial x_3} \end{bmatrix}$$

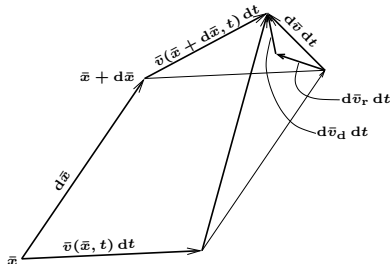
$$\bar{\bar{T}}_r = \frac{1}{2} \begin{bmatrix} 0 & \left(\frac{\partial v_2}{\partial x_1} - \frac{\partial v_1}{\partial x_2} \right) & \left(\frac{\partial v_3}{\partial x_1} - \frac{\partial v_1}{\partial x_3} \right) \\ - \left(\frac{\partial v_2}{\partial x_1} - \frac{\partial v_1}{\partial x_2} \right) & 0 & \left(\frac{\partial v_3}{\partial x_2} - \frac{\partial v_2}{\partial x_3} \right) \\ - \left(\frac{\partial v_3}{\partial x_1} - \frac{\partial v_1}{\partial x_3} \right) & - \left(\frac{\partial v_3}{\partial x_2} - \frac{\partial v_2}{\partial x_3} \right) & 0 \end{bmatrix} = \frac{1}{2} \begin{bmatrix} 0 & \omega_3 & -\omega_2 \\ -\omega_3 & 0 & \omega_1 \\ \omega_2 & -\omega_1 & 0 \end{bmatrix}$$

where ω_1 , ω_2 and ω_3 are the three components of the vorticity vector $\bar{\omega} = \nabla \wedge \bar{v}$

Relative motion near a point

$$\nabla \bar{v} = \frac{1}{2}(\nabla \bar{v} + \nabla \bar{v}^T) + \frac{1}{2}(\nabla \bar{v} - \nabla \bar{v}^T) = \bar{\mathbf{T}}_d + \bar{\mathbf{T}}_r$$

$$d\bar{v} = d\bar{x} \cdot \nabla \bar{v} = d\bar{x} \cdot \bar{\mathbf{T}}_d + d\bar{x} \cdot \bar{\mathbf{T}}_r = d\bar{v}_d + d\bar{v}_r$$



If $\bar{\mathbf{T}}_d$ were identically zero, the local motion of the fluid would be exactly that of a rigid body (translation + rotation)

rotation rate

$$d\bar{v}_r = d\bar{x} \cdot \bar{\mathbf{T}}_r = \frac{1}{2}(\nabla \wedge \bar{v}) \wedge d\bar{x} = \frac{1}{2}\bar{\omega} \wedge d\bar{x}$$

→ rotation of the element $d\bar{x}$ with angular velocity $\frac{\bar{\omega}}{2}$

deformation rate or strain rate

$$d\bar{v}_d = d\bar{x} \cdot \bar{\mathbf{T}}_d = \bar{\mathbf{T}}_d \cdot d\bar{x} = \bar{\mathbf{T}}_d \cdot \bar{n} ds$$

ds : length of the element $d\bar{x}$

$\bar{n}$: unit vector in the direction of $d\bar{x}$

extension: $\bar{n} \cdot \bar{\mathbf{T}}_d \cdot \bar{n} ds$

shear strain: $[\bar{\mathbf{T}}_d \cdot \bar{n} - (\bar{n} \cdot \bar{\mathbf{T}}_d \cdot \bar{n})\bar{n}] ds$

Principal directions of strain

strain rate p.u.length: $d\bar{v}_d = \bar{\bar{T}}_d \cdot \bar{n}$

In general, the strain rate is not aligned with $\bar{n}$:

extension: $\bar{n} \cdot \bar{\bar{T}}_d \cdot \bar{n}$

shear strain: $[\bar{\bar{T}}_d \cdot \bar{n} - (\bar{n} \cdot \bar{\bar{T}}_d \cdot \bar{n})\bar{n}]$

principal directions of strain

Three privileged directions of space along which the strain reduces to an extension, without any shear strain (the resulting vector $d\bar{v}_d$ is parallel to $\bar{n}$)

- These principal directions $\bar{n}_1$, $\bar{n}_2$ and $\bar{n}_3$, and their corresponding strain rates, λ_1 , λ_2 and λ_3 , are computed by solving the eigenvalue problem

$$\bar{\bar{T}}_d \cdot \bar{n} = \lambda \bar{n}$$

- In particular, the characteristic equation that determines λ_i ,

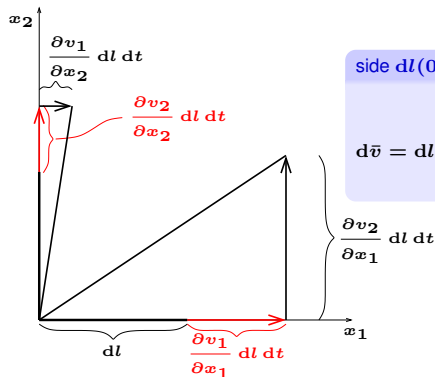
$$|\bar{\bar{T}}_d - \lambda \bar{\bar{I}}| = 0$$

follows from imposing the existence of nontrivial solutions.

- The symmetry of the tensor $\bar{\bar{T}}_d$ guarantees the existence of three real roots for this equation, associated with three mutually perpendicular principal directions.
- In the local reference frame defined by the three principal directions $\bar{n}_i$ the associated rate-of-strain tensor becomes diagonal, with diagonal components λ_1 , λ_2 and λ_3 .

Deformation of a square fluid element

$$d\bar{l} \Rightarrow d\bar{l} + d\bar{v} dt = d\bar{l} + d\bar{l} \cdot \nabla \bar{v} dt$$



side $dl(0, 1)$

$$d\bar{v} = dl(0, 1) \cdot \begin{bmatrix} \frac{\partial v_1}{\partial x_1} & \frac{\partial v_2}{\partial x_1} \\ \frac{\partial v_1}{\partial x_2} & \frac{\partial v_2}{\partial x_2} \end{bmatrix} = dl \left(\frac{\partial v_1}{\partial x_2}, \frac{\partial v_2}{\partial x_2} \right)$$

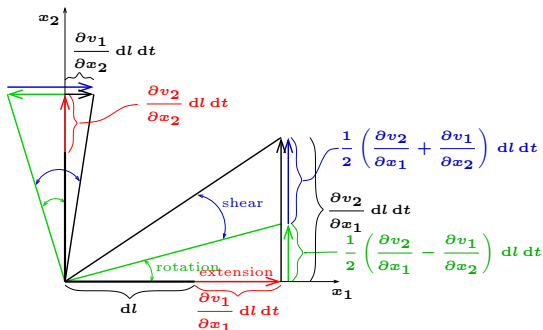
side $dl(1, 0)$

$$d\bar{v} = dl(1, 0) \cdot \begin{bmatrix} \frac{\partial v_1}{\partial x_1} & \frac{\partial v_2}{\partial x_1} \\ \frac{\partial v_1}{\partial x_2} & \frac{\partial v_2}{\partial x_2} \end{bmatrix} = dl \left(\frac{\partial v_1}{\partial x_1}, \frac{\partial v_2}{\partial x_1} \right)$$

Deformation of a square fluid element, decomposing the motion

$$\bar{\bar{\mathbf{T}}}_d = \begin{bmatrix} \frac{\partial v_1}{\partial x_1} & \frac{1}{2} \left(\frac{\partial v_2}{\partial x_1} + \frac{\partial v_1}{\partial x_2} \right) \\ \frac{1}{2} \left(\frac{\partial v_2}{\partial x_1} + \frac{\partial v_1}{\partial x_2} \right) & \frac{\partial v_2}{\partial x_2} \end{bmatrix} \quad \bar{\bar{\mathbf{T}}}_r = \begin{bmatrix} 0 & \frac{1}{2} \left(\frac{\partial v_2}{\partial x_1} - \frac{\partial v_1}{\partial x_2} \right) \\ -\frac{1}{2} \left(\frac{\partial v_2}{\partial x_1} - \frac{\partial v_1}{\partial x_2} \right) & 0 \end{bmatrix}$$

$$d\bar{l} \cdot \nabla \bar{v} dt = \underbrace{d\bar{l} \cdot \bar{\bar{\mathbf{T}}}_d}_{d\bar{v}_d} dt + \underbrace{d\bar{l} \cdot \bar{\bar{\mathbf{T}}}_r}_{d\bar{v}_r} dt$$



side $d\mathbf{l}(1, 0)$

$$d\bar{v}_d = d\mathbf{l} \left[\frac{\partial v_1}{\partial x_1}, \frac{1}{2} \left(\frac{\partial v_2}{\partial x_1} + \frac{\partial v_1}{\partial x_2} \right) \right]$$

$$d\bar{v}_r = d\mathbf{l} \left[0, \frac{1}{2} \left(\frac{\partial v_2}{\partial x_1} - \frac{\partial v_1}{\partial x_2} \right) \right]$$

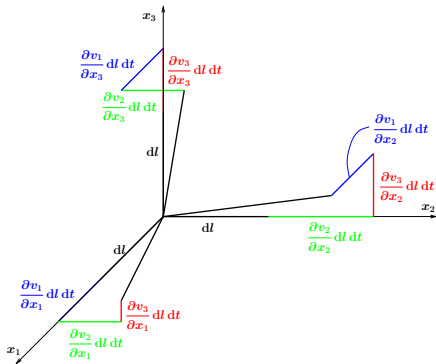
side $d\mathbf{l}(0, 1)$

$$d\bar{v}_d = d\mathbf{l} \left[\frac{1}{2} \left(\frac{\partial v_2}{\partial x_1} + \frac{\partial v_1}{\partial x_2} \right), \frac{\partial v_2}{\partial x_2} \right]$$

$$d\bar{v}_r = d\mathbf{l} \left[-\frac{1}{2} \left(\frac{\partial v_2}{\partial x_1} - \frac{\partial v_1}{\partial x_2} \right), 0 \right]$$

Deformation of a cubic fluid element

$$d\bar{l} \Rightarrow d\bar{l} + \underbrace{d\bar{l} \cdot \nabla \bar{v}}_{d\bar{v}} dt$$



side $dl(1, 0, 0)$

$$d\bar{v} = dl(1, 0, 0) \cdot \begin{bmatrix} \frac{\partial v_1}{\partial x_1} & \frac{\partial v_2}{\partial x_1} & \frac{\partial v_3}{\partial x_1} \\ \frac{\partial v_1}{\partial x_2} & \frac{\partial v_2}{\partial x_2} & \frac{\partial v_3}{\partial x_2} \\ \frac{\partial v_1}{\partial x_3} & \frac{\partial v_2}{\partial x_3} & \frac{\partial v_3}{\partial x_3} \end{bmatrix}$$

$$dl(1, 0, 0) \Rightarrow d\bar{l} + dl dt \left(\frac{\partial v_1}{\partial x_1}, \frac{\partial v_2}{\partial x_1}, \frac{\partial v_3}{\partial x_1} \right)$$

$$\Rightarrow dl \left(1 + \frac{\partial v_1}{\partial x_1} dt, \frac{\partial v_2}{\partial x_1} dt, \frac{\partial v_3}{\partial x_1} dt \right)$$

side $dl(0, 1, 0)$

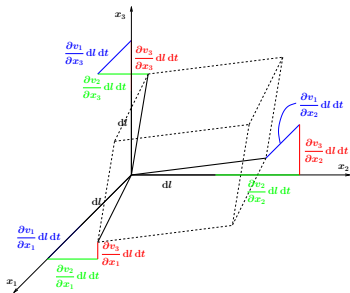
$$\Rightarrow dl \left(\frac{\partial v_1}{\partial x_2} dt, 1 + \frac{\partial v_2}{\partial x_2} dt, \frac{\partial v_3}{\partial x_2} dt \right)$$

side $dl(0, 0, 1)$

$$\Rightarrow dl \left(\frac{\partial v_1}{\partial x_3} dt, \frac{\partial v_2}{\partial x_3} dt, 1 + \frac{\partial v_3}{\partial x_3} dt \right)$$

Variation of volume

The distortion of the fluid element associated with the straining motion produces a variation of volume from its initial value dl^3 .



$$V_f = dl^3 \begin{vmatrix} 1 + \frac{\partial v_1}{\partial x_1} dt & \frac{\partial v_2}{\partial x_1} dt & \frac{\partial v_3}{\partial x_1} dt \\ \frac{\partial v_1}{\partial x_2} dt & 1 + \frac{\partial v_2}{\partial x_2} dt & \frac{\partial v_3}{\partial x_2} dt \\ \frac{\partial v_1}{\partial x_3} dt & \frac{\partial v_2}{\partial x_3} dt & 1 + \frac{\partial v_3}{\partial x_3} dt \end{vmatrix}$$

$$= dl^3 + dl^3 dt \left(\frac{\partial v_1}{\partial x_1} + \frac{\partial v_2}{\partial x_2} + \frac{\partial v_3}{\partial x_3} \right),$$

where small terms of order dt^2 and dt^3 are neglected.

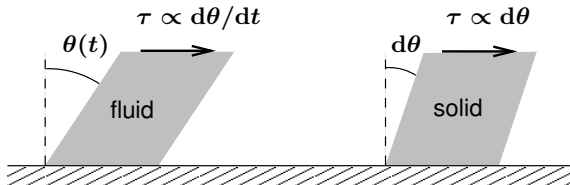
rate of expansion

$$\frac{1}{V} \frac{dV}{dt} = \frac{1}{dt} \frac{V_f - V_0}{V_0} = \frac{\partial v_1}{\partial x_1} + \frac{\partial v_2}{\partial x_2} + \frac{\partial v_3}{\partial x_3} = \nabla \cdot \vec{v}$$

for a perfect liquid (constant density)

$$\nabla \cdot \vec{v} = 0$$

Figures from the notes

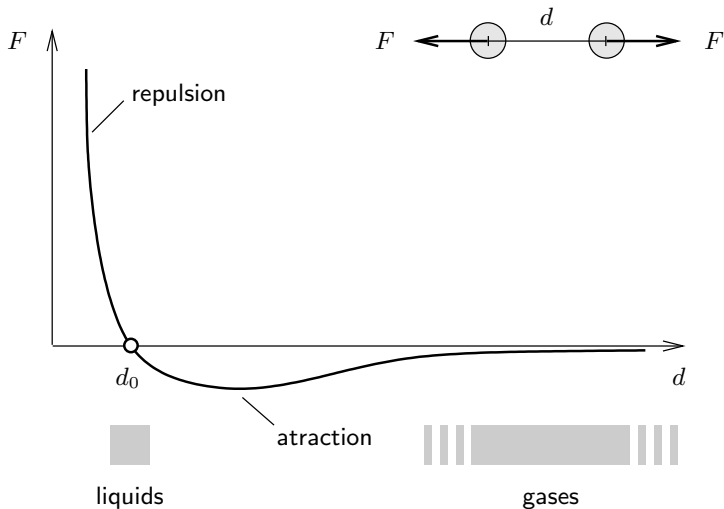


Under the action of a small external force, solids respond with a small deformation. In solids, the internal force counteracting the external force is proportional to the deformation. Fluids are easily deformed under the action of external forces. The internal resistive force turns out to be proportional to the rate of deformation, rather than to the deformation itself.

Under the action of a tangential external force, fluids continuously deform...

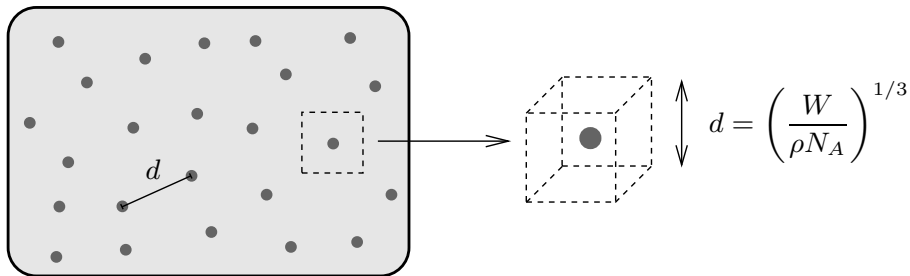


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Schematic representation of the force between two molecules as a function of the distance between their centers.

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In average, the volume occupied by one molecule is a cube of side d , d being the average intermolecular distance. Knowing the fluid density ρ and its molar mass W it is straightforward to estimate the average distance d between two molecules according to

$$\rho = \frac{W/N_A}{d^3} \equiv \frac{\text{mass of 1 molecule}}{\text{volume occupied by 1 molecule}} \rightarrow d = \left(\frac{W}{\rho N_A} \right)^{1/3}$$

(return)

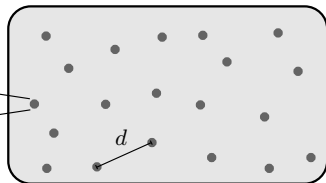
Gases ($d \simeq 10 d_0$)

Low density

High compressibility

Use all the volume available

$$d_0 \simeq 3 - 4 \text{ \AA}$$

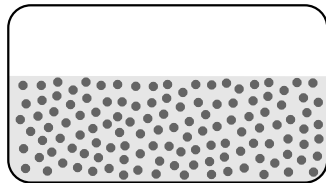


Liquids ($d \simeq d_0$)

High density

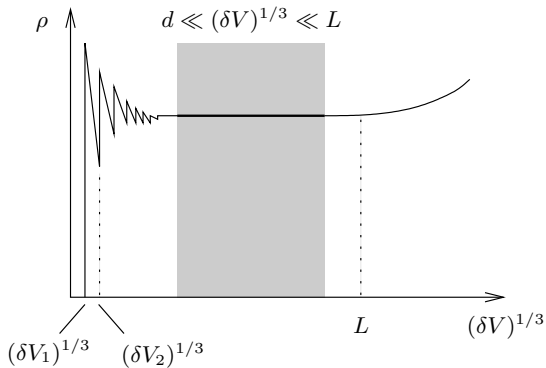
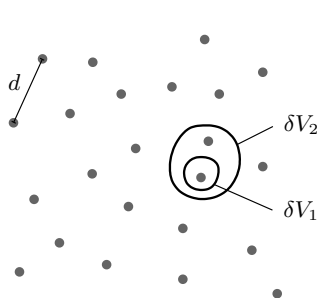
Low compressibility

Form free surfaces



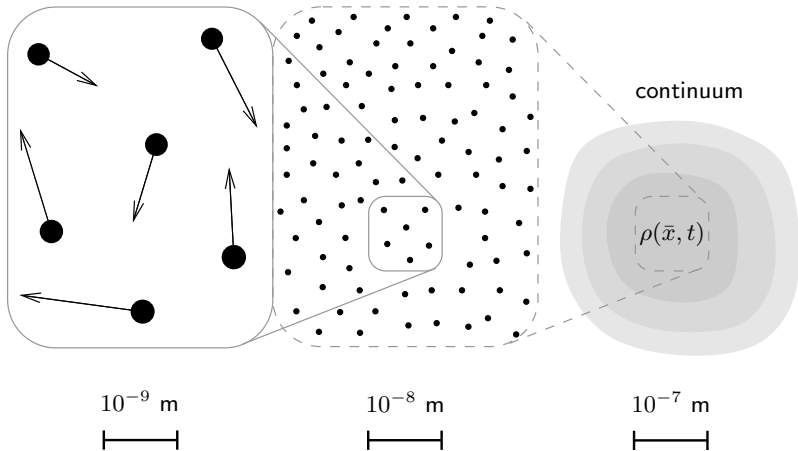
The differences in macroscopic properties of liquids and gases are the result of the different microscopic structure.

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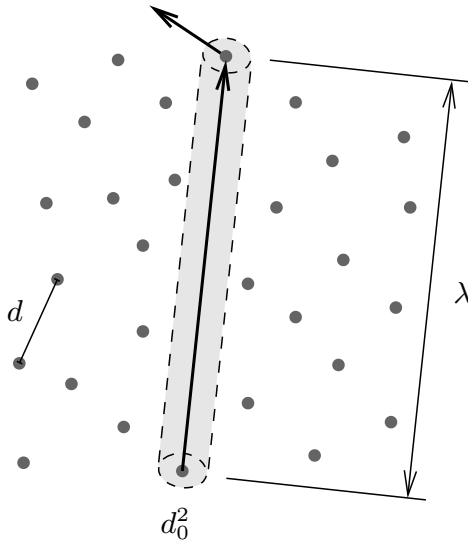
The concept of fluid particle.

(return)



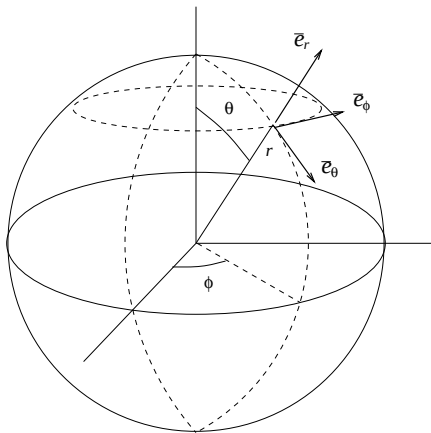
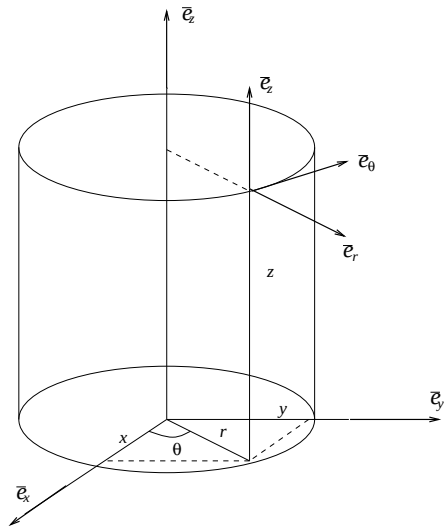
The continuum hypothesis, $d/L \ll 1$, allows to define a range of scales between the microscopic characteristic scale, d , and the macroscopic characteristic scale, L , where all fluid properties can be described as continuous functions of position $\bar{x}$ and time t in terms of the properties of each fluid particle.

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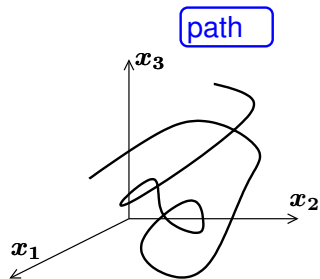
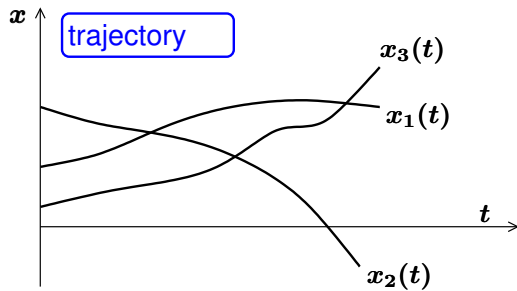
The average distance travelled by a molecule between two subsequent collisions, called the *mean free path* λ , can be estimated by equating the volume swept by the molecule in its displacement between collisions, $d_0^2 \lambda$, to the volume per molecule in the gas, d^3 , to give $\lambda/d \simeq (d/d_0)^2$.

(return)



Schematic representation of relevant coordinate systems

(return)



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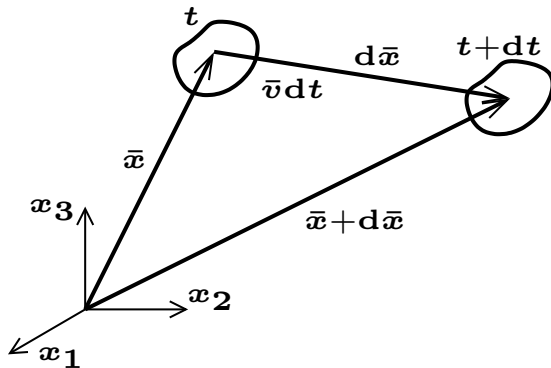


Figure 2.2: Infinitesimal displacement of a fluid particle

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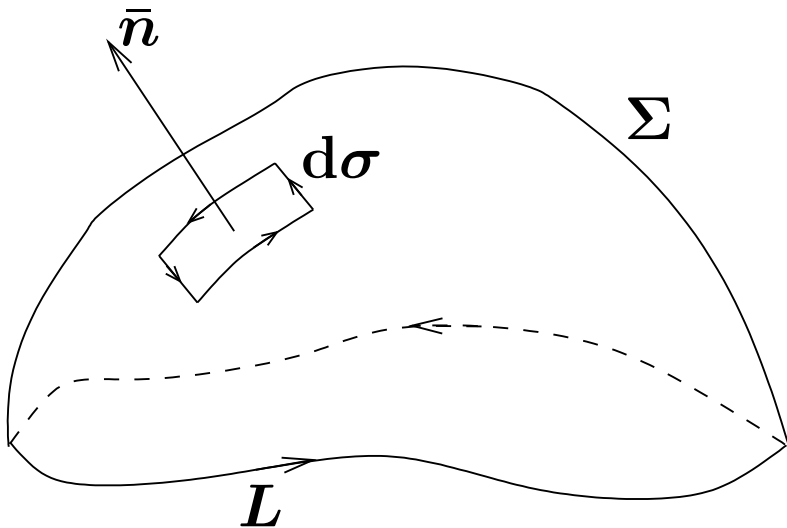


Figure 2.3: Application of Stokes theorem for the computation of the circulation.

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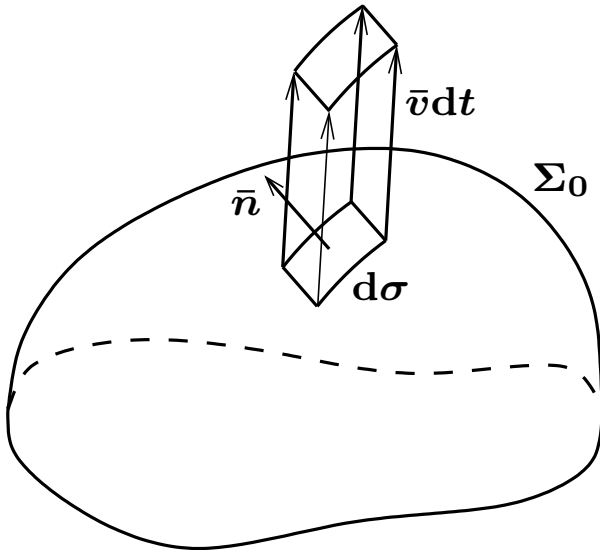


Figure 2.4: Convective flux.

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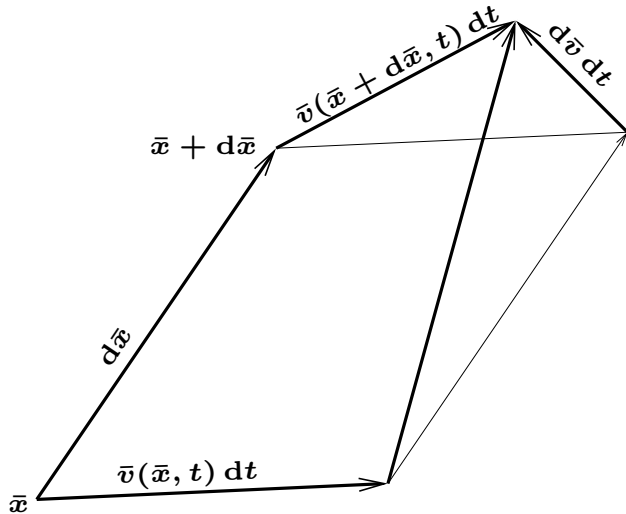


Figure 2.6: Motion of a differential element of fluid line.

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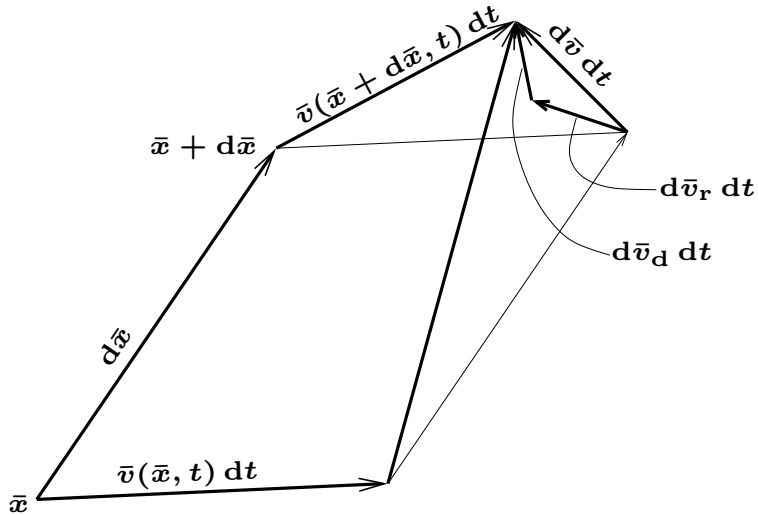


Figure 2.6: Motion of a differential element of fluid line.

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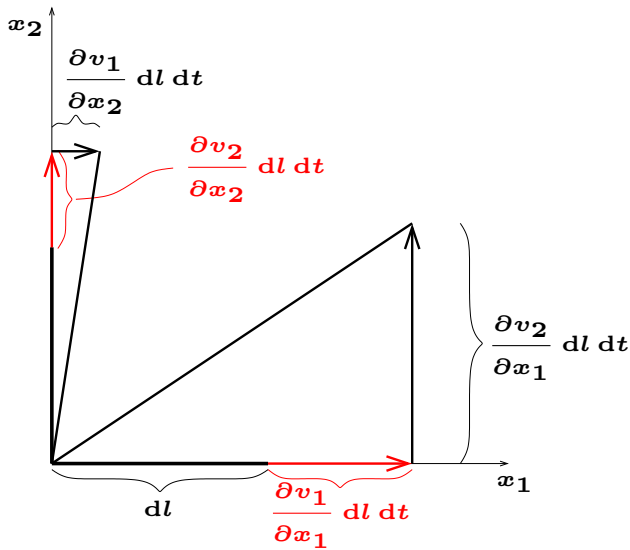


Figure 2.7: Deformation of an infinitesimally small fluid element of square shape.

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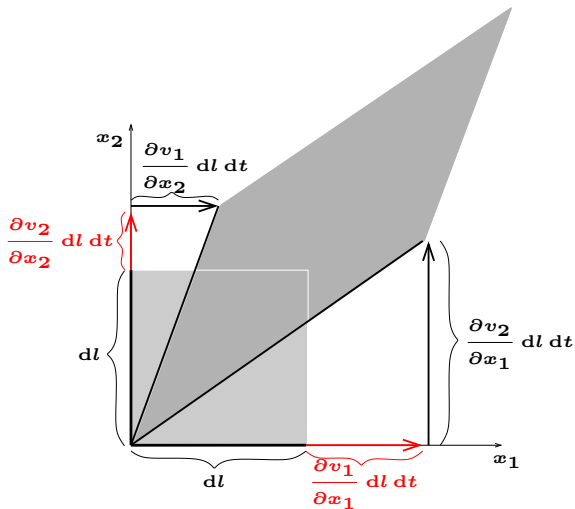


Figure 2.7: Deformation of an infinitesimally small fluid element of square shape.

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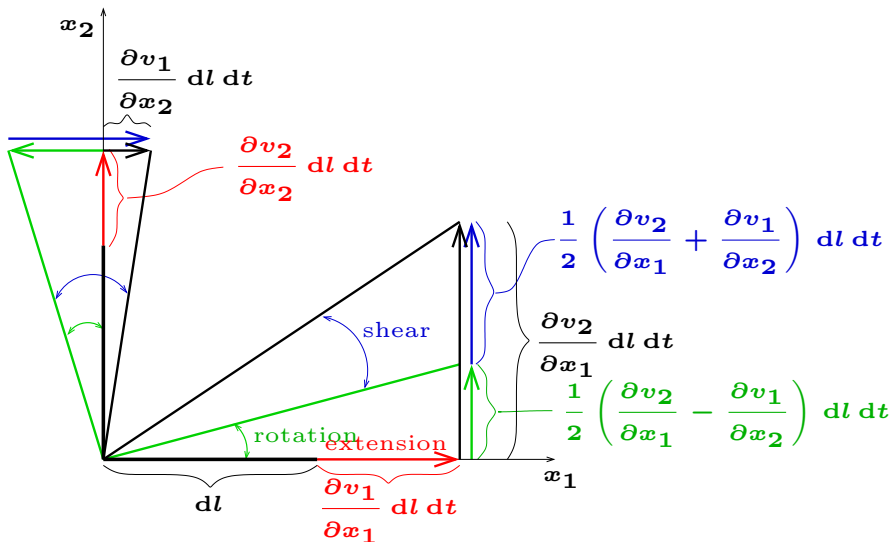


Figure 2.7: Rotation and shear of an infinitesimally small fluid element of square shape.

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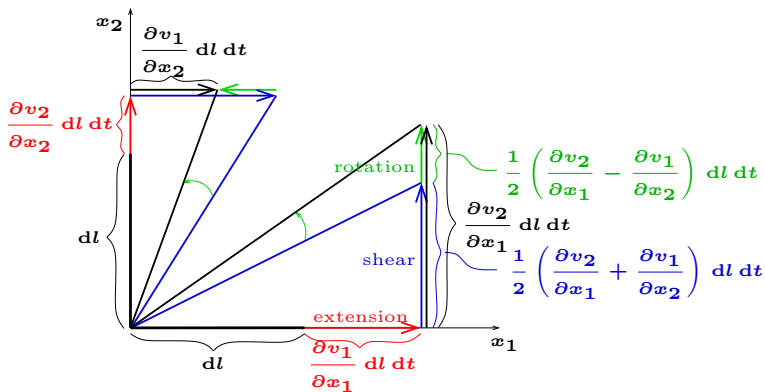


Figure 2.7: Rotation and shear of an infinitesimally small fluid element of square shape.

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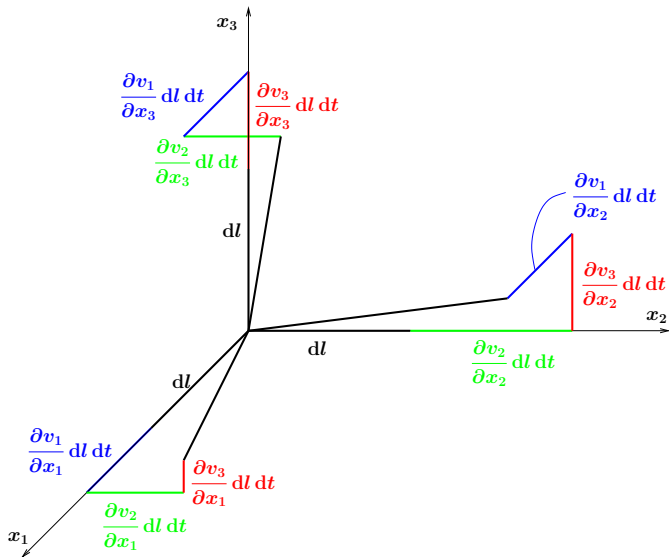


Figure 2.8: Deformation and rotation of a cubic fluid element of infinitesimally small size.

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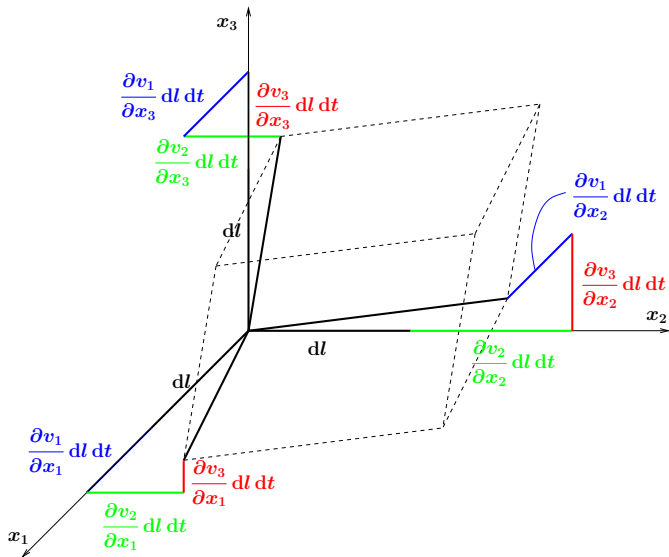


Figure 2.8: Deformation and rotation of a cubic fluid element of infinitesimally small size.

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